

**BS COMPUTER SCIENCE
FINAL YEAR PROJECT DOCUMENTATION**

AI LifeSaver

AI-Powered First Aid and Emergency Guidance Platform

Full Title: AI LifeSaver for Edge-Accelerated Pre-Hospital Emergency Guidance

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Abstract

Pre-hospital emergency mortality across Pakistan's metropolitan centers is driven by an acute intervention vacuum during the first ten minutes following traumatic injury or cardiac collapse. Clinical data from Karachi reveals that over fifty-eight percent of fatal trauma victims succumb to their injuries prior to reaching hospital facilities, while out-of-hospital cardiac arrest survival remains under two percent. Although more than ninety percent of these catastrophic incidents are witnessed by family members or bystanders, bystander cardiopulmonary resuscitation is initiated in only 2.3 percent of cases. Untrained individuals are frequently paralyzed by panic or perform contraindicated actions — such as administering liquids to comatose patients, applying harmful domestic substances to acute thermal burns, or forcibly moving unstable cervical spines — that directly accelerate fatal complications.

AI LifeSaver resolves this pre-hospital crisis by converting consumer mobile and desktop browsers into deterministic, bilingual emergency triage terminals without requiring native application installation. The platform combines high-throughput, edge-orchestrated Large Language Model inference via Groq-hosted LLaMA 3.3 70B and Google Gemini engines with a verified relational first-aid repository managed on Supabase PostgreSQL. Operating under a strict Retrieval-Augmented Generation paradigm, the system eliminates medical hallucinations and delivers bilingual (English and Urdu) step-by-step guidance, audio narration, and one-touch integration with emergency dispatch services.

To ensure real-world viability across bandwidth-constrained environments, the platform implements a hybrid failover pipeline. High-speed conversational triage is processed through Supabase Edge Functions at sub-second latencies, while static procedural records, Web Speech synthesis pipelines, and client-side vector search run entirely locally when connectivity is lost. By pairing algorithmic safety guardrails with immediate one-touch dispatch links to services like Sindh Integrated Emergency & Health Services (SIEHS) Rescue 1122 and the Edhi Foundation, AI LifeSaver bridges the life-critical gap between symptom onset and professional medical response.

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Chapter 1: Project Overview

AI LifeSaver is an edge-accelerated, bilingual emergency first-aid assistance platform engineered to stabilize victims of acute trauma and life-threatening medical events during the pre-hospital interval. Within the emergency medicine timeline, the period immediately following an incident — frequently termed the "golden window" — determines long-term morbidity and mortality. When professional Emergency Medical Services (EMS) experience protracted transit times due to urban traffic density or decentralized dispatch infrastructure, on-scene bystanders represent the only operational resource capable of sustaining life.

The platform addresses this critical window through an accessible web architecture that requires no account registration, financial subscription, or app-store installation. Developed within the Department of Computer Science at the University of Karachi, AI LifeSaver provides culturally grounded, linguistically localized medical instructions in English and Urdu. By combining sub-second artificial intelligence inference with deterministic local clinical datasets, the platform guides laypersons through complex emergency protocols — such as chest compressions, severe hemorrhage packing, foreign body airway clearance, and burn cooling — while maintaining a continuous bridge to physical emergency dispatch.

Chapter 2: Problem Statement

The pre-hospital emergency care infrastructure across developing nations, and specifically throughout urban Pakistan, suffers from systemic delays that contribute directly to elevated casualty rates. Epidemiological analyses demonstrate that the baseline survivability for critical medical emergencies in Pakistan remains significantly below international clinical standards.

Clinical / Operational Metric	Pakistan Urban Benchmark (Karachi)	International Standard Baseline	Source
Bystander CPR Initiation Rate	2.3% of witnessed cardiac events	40.0% – 55.0% in established EMS registries	BMC Emergency Medicine (Khan et al.)
Pre-Hospital Trauma Mortality	58.0% of violent trauma deaths occur pre-ED	< 20.0% under integrated trauma networks	Journal of Pakistan Medical Assoc.
Dead on Arrival (DOA) Burden	1.0 to 2.0 per 1,000 emergency admissions	< 0.2 per 1,000 emergency admissions	Resuscitation Outcomes Consortium
Median Trauma Transit Latency	30.0 minutes (IQR: 20–40 minutes)	< 8.0 minutes to advanced life support	PMC Emergency Trauma Cohort
Predominant Transit Methodology	80.2% basic transport without life support	> 90.0% Advanced Life Support (ALS) units	Aga Khan University Emergency Medicine

Table 2.1 — Pakistan pre-hospital care benchmarks vs. international standards

The clinical consequences of these delays are compounded by three interconnected failures in the field:

The Interventional Vacuum: Acute arterial lacerations cause lethal exsanguination within three to five minutes, and global cerebral ischemia causes irreversible cortical necrosis within four to six minutes. When ambulance transit times extend beyond twenty minutes, bystander inaction guarantees biological death prior to EMS arrival.

Layperson Panic and Decision Paralysis: Untrained bystanders at accident scenes experience acute panic, leading to decision paralysis. Relatives frequently watch casualties deteriorate without attempting basic airway clearance or circulatory support.

Harmful Contraindicated Interventions: Untrained individuals routinely perform dangerous layman practices. These include administering oral water or milk to unconscious patients (inducing pulmonary aspiration), forcibly moving polytrauma casualties without cervical stabilization (causing permanent spinal cord severance), applying toothpaste, grease, or raw flour to severe thermal burns (exacerbating tissue necrosis and sepsis), and utilizing uncalibrated wire tourniquets (causing ischemic limb loss).

Linguistic and Interface Barriers: Most emergency medical applications rely entirely on complex, English-language terminology with dense text hierarchies, rendering them unusable for the broader Urdu-speaking public during high-stress scenarios.

Chapter 3: Project Objectives

The engineering design of AI LifeSaver balances distributed systems performance with strict clinical determinism:

1. **Sub-Second Generative Inference:** Implement an artificial intelligence inference pipeline capable of a Time-to-First-Token (TTFT) under 1,200 milliseconds by orchestrating Groq-hosted LLaMA 3.3 70B models through edge-deployed serverless proxies.
2. **Zero-Hallucination Retrieval Architecture:** Bind generative language models to verified, relational clinical records stored in Supabase PostgreSQL, ensuring the complete elimination of synthetic medical claims or unverified pharmaceutical advice.
3. **Full Bilingual Parity:** Deliver complete bidirectional localization (English and Urdu) featuring dynamic Right-to-Left (RTL) document re-orientation, culturally attuned medical vocabulary, and phonetic matching for Roman Urdu search queries.
4. **Hands-Free Auditory Pipeline:** Provide multimodal speech input via the Web Speech API and synchronized text-to-speech auditory loops, allowing responders to receive real-time spoken guidance without looking away from the patient.
5. **Deterministic Offline Operability:** Maintain complete core emergency search and step-by-step guidance capabilities during complete cellular or broadband outages via client-side caching and dynamic client-rendered PDF guides.

6. **Emergency Telephony Integration:** Provide immediate, one-touch telephony bridging to localized emergency infrastructure, including SIEHS Rescue 1122 and Edhi 115, alongside geospatial emergency facility mapping via OpenStreetMap.

Chapter 4: Proposed Solution

AI LifeSaver functions as a client-side Single Page Application (SPA) linked to an edge-compute layer and a relational database. Instead of relying on monolithic servers, the platform uses an edge-first, decoupled architecture.

Stage	User Action / Trigger	Underlying Mechanism	Output State
1. Emergency Ingress	User accesses web URL via browser; inputs search via speech or text.	Vite 7 SPA bundle hydrates locally; Web Speech API captures audio buffers.	Immediate interactive view; low-contrast elements suppressed.
2. Triage & Retrieval	Input parsed for clinical entities, symptoms, or injury types.	Trigram index (pg_trgm) queries PostgreSQL condition catalog directly.	Instant display of categorized condition cards with risk badges.
3. Conversational AI	Complex, multi-symptom, or ambiguous emergency prompt submitted.	Edge Function enriches query with matched protocols; executes Groq LPU inference.	Real-time SSE token stream rendering conversational advice.
4. Guided Stabilization	Bystander selects specific condition protocol card.	Client loads step-by-step procedural array; activates Web Speech Synthesis loop.	Visual action steps rendered alongside sequential voice guidance.
5. Physical Escalation	Bystander clicks SOS dialer or inspects nearby trauma facilities.	Native tel: URI invokes OS dialer; Leaflet maps spatial coordinates via OpenStreetMap.	Direct voice call to 1122 initiated; interactive hospital map displayed.
6. Disconnected State	Mobile connection drops due to infrastructure outage.	Service logic falls back to bundled firstAid.json; jsPDF renders offline documents.	Uninterrupted first-aid execution; offline indicator surfaced.

Table 4.1 — End-to-end operational pipeline

Chapter 5: Scope of the Project

5.1 In-Scope Features

The implemented architecture of AI LifeSaver delivers the following concrete technical deliverables:

- **Voice-Driven Symptom Search:** Speech recognition interface capable of parsing English and Roman Urdu spoken queries into standardized search tokens.
- **Conversational First-Aid Assistant:** Serverless edge-proxied AI engine powered by Groq LLaMA 3.3 with automated fallback to Google Gemini 1.5 Flash.

- **Visual Triage Cards:** Standardized emergency action cards featuring danger callouts, numbered execution steps, and explicit contraindication warnings.
- **Auditory Step Playback:** Integrated browser speech synthesis engine that vocally reads emergency instructions sequentially in English and Urdu.
- **Full Bilingual Interface:** Dynamic global state toggle managing complete typography, button labels, RTL orientation, and localized condition records.
- **Emergency Dispatch Hotlines:** One-touch direct dialing anchors for primary Pakistani emergency networks, specifically SIEHS Rescue 1122, Edhi 115, and Chhipa.
- **Interactive Facility Map:** Leaflet and OpenStreetMap integration rendering geographical locations, routes, and contact numbers for nearby hospitals and blood banks.
- **Offline PDF Generator:** Client-side document generation engine using jsPDF that converts procedural data into printable emergency reference sheets.

5.2 Out-of-Scope for the Initial MVP

To maintain clinical safety and software reliability, the following systems are strictly excluded from this release:

- **Municipal CAD Direct Integration:** Automated insertion of incident tickets into public ambulance computer-aided dispatch networks.
- **Computer-Vision Diagnostics:** Automated classification of wound depths, fracture angulation, or burn percentages using device camera streams.
- **Biometric Signal Telemetry:** Real-time pairing with consumer smartwatches to read photoplethysmography or electrocardiogram streams.
- **Invasive Medical Procedures:** Instructions detailing intravenous cannulation, surgical cricothyroidotomy, needle thoracostomy, or prescription pharmacology.
- **Authenticated User Profiling:** Mandatory login walls, personal medical record storage, or cloud tracking of sensitive personal health records.

Chapter 6: Functional Requirements

6.1 Authentication and Guest State Session Management

In life-critical emergency contexts, identity verification requirements introduce unacceptable latency:

- **Zero-Friction Access:** The application must grant unauthenticated guest access to all search systems, emergency procedure guides, conversational assistants, and emergency contact directories.
- **Stateless Client Persistence:** User interface preferences — including language selection, speech synthesis velocity, and volume thresholds — must be persisted exclusively in browser localStorage without generating server-side user tracking records.

6.2 Bilingual Localization Engine (English / Urdu RTL)

The platform must provide linguistic parity between English and Urdu across all presentation surfaces:

- **Directional Document Transformation:** When switching between English and Urdu, the system must dynamically reconfigure the root HTML Document Object Model attributes (`dir="rtl"` or `dir="ltr"` and `lang="ur"` or `lang="en"`) to adjust text alignment, margins, and flex-direction vectors.
- **Synchronized Translation Catalog:** All interface controls, navigation links, error messages, and dynamic state descriptions must be driven by a structured translation catalog (`texts.json`).
- **Parallel Medical Datasets:** Relational condition tables must store dual-language string attributes for clinical names, summaries, step instructions, and danger warnings.

6.3 Multimodal Emergency Search and Keyword Indexing

The retrieval system must resolve ambiguous or noisy user inputs into standardized clinical protocols:

- **Phonetic and Trigram String Matching:** The search layer must match user queries against condition titles, medical terms, and vernacular synonyms using PostgreSQL Generalized Index Search (`gin_trgm_ops`).
- **Web Speech Audio Capture:** The interface must capture spoken queries via the browser's native SpeechRecognition API, parse the resulting audio buffers into text, and execute searches without manual keyboard input.
- **Visual Urgency Badging:** Retrieved search results must immediately display color-coded risk indicators reflecting their medical severity.

6.4 Real-Time Conversational AI First Aid Assistant

The conversational subsystem must handle complex natural language queries that fall outside exact keyword matches:

- **Edge Proxy Intermediation:** Client chat requests must route through a secure Supabase Edge Function to prevent public exposure of inference API keys.
- **Contextual Grounding (RAG):** The Edge Function must search the internal database for relevant condition records and inject these verified steps directly into the model's system prompt.
- **Token Streaming:** The assistant must stream responses incrementally to the client using Server-Sent Events, ensuring users receive actionable instructions without waiting for the full response to generate.

6.5 Step-by-Step Audio Narration and Hands-Free Protocol Guidance

The audio subsystem must deliver hands-free auditory instructions while a bystander treats a patient:

- **Synthesized Voice Generation:** The system must use the W3C Web Speech Synthesis API to read the currently selected procedural step aloud.

- **Linguistic Voice Resolution:** The audio engine must detect the active language state and bind to localized voice packages (en-US for English, ur-PK for Urdu) supported by the client operating system.
- **Sequential Playback Controls:** The user interface must provide large, high-contrast controls to play, pause, repeat, or step through instructions sequentially.

6.6 Geolocation Directory and One-Touch Emergency SOS Telephony

The platform must connect digital first-aid guidance with physical pre-hospital infrastructure:

- **One-Touch Telephony Anchors:** The interface must provide prominent SOS buttons that use native tel: URIs to open the device dialer with regional emergency numbers (Rescue 1122, Edhi 115) pre-loaded.
- **Spatial Facility Visualization:** The system must render an interactive Leaflet map marking public trauma centers, specialized burn units, and blood banks using OpenStreetMap tile servers.
- **Fallback Directory Tables:** If geospatial mapping APIs fail to load, the system must display a pre-compiled, static contact directory of major hospitals and ambulance dispatch centers.

6.7 Offline Resilience and Portable Document Export

The application must remain functional during complete broadband or cellular network disruptions:

- **Static Dataset Failover:** If network requests to Supabase fail, the client data layer must automatically fall back to an internal JSON dataset (firstAid.json) containing core emergency procedures.
- **Client-Side Document Compilation:** The application must compile and download formatted, printable emergency guides locally using jsPDF without requiring server-side rendering.

Chapter 7: User Roles

AI LifeSaver enforces a role-based access model using Supabase Row Level Security (RLS) to protect clinical data while maintaining friction-free emergency access.

Role	Authentication Method	Clinical Access Privileges	Administrative Authority
Unauthenticated Bystander	Public Anon Key (anon-key)	Unrestricted read access to all condition records, procedures, mapping coordinates, and the streaming AI assistant.	Strictly none; cannot modify, append, or delete any remote data records.
Community Responder / Student	Public Anon Key (anon-key)	Unrestricted read access; local offline data caching; client-side PDF guide generation.	Strictly none; purely consumer-tier execution.
Clinical Content Administrator	Supabase Auth (JWT with admin role)	Full read, write, update, and delete access across all condition schemas,	Complete schema administration, API

Role	Authentication Method	Clinical Access Privileges	Administrative Authority
		steps, synonyms, and emergency directories.	key rotation, edge function deployment, and system prompt modification.

Table 7.1 — Role-based access control matrix

Chapter 8: Technology Stack

The software stack for AI LifeSaver was selected to achieve high edge throughput, minimal bundle overhead, and zero-latency local execution.

Layer	Selected Tool	Alternatives Evaluated	Justification
Client Application	React 19 + Vite 7	Next.js 14 / Nuxt.js	Eliminates server-side rendering overhead; compiles into a lightweight static SPA that executes client-side during outages.
Styling & Layout	Tailwind CSS v4	Bootstrap 5 / Vanilla CSS	High-performance atomic utility classes, minimal production CSS bundle footprints, and native support for RTL variants.
Animation Engine	Framer Motion	CSS Keyframes	Provides fluid, hardware-accelerated transitions between triage steps without blocking the JavaScript main thread.
Data Layer & Search	Supabase PostgreSQL 15	MongoDB Atlas / Firebase	Native relational integrity, built-in Row Level Security, and high-speed trigram text indexing via the <code>pg_trgm</code> extension.
Serverless Compute	Supabase Edge Functions (Deno)	AWS Lambda / Cloudflare	Globally distributed Deno runtime providing low-latency API proxying and streaming responses with minimal cold-start times.
Primary AI Engine	Groq LPU API (LLaMA 3.3 70B)	OpenAI GPT-4o / Claude 3.5	Groq's specialized inference hardware achieves sub-second token generation, essential for high-stress emergency guidance.
Secondary AI Engine	Google Gemini 1.5 Flash	Anthropic Claude 3 Haiku	Serves as an automated fallback engine, offering high throughput and stable failover if primary API rate limits occur.

Layer	Selected Tool	Alternatives Evaluated	Justification
Geospatial Mapping	Leaflet + OpenStreetMap	Google Maps JavaScript API	Open-source foundation requires no proprietary API keys, has a smaller bundle size, and operates under permissive terms.
Audio I/O Processing	W3C Web Speech API	AWS Polly / Whisper API	Executes natively within the browser, eliminating external network latency and third-party processing costs.
Document Export	jsPDF	Puppeteer / Server PDF	Generates printable vector PDF documents directly in the client browser without needing external backend rendering.

Table 8.1 — Technology stack and rationale

Chapter 9: Frontend Design

9.1 Main Pages

The frontend interface is organized into focused, single-purpose views to minimize cognitive load during crises:

- **Home and Emergency Hub (Home.jsx):** Serves as the primary emergency entry point. Features a prominent search bar, one-touch voice recognition triggers, and immediate-action buttons for common critical emergencies: Cardiopulmonary Arrest, Severe Hemorrhage, Thermal Burns, and Choking.
- **Procedural Emergency Viewer (Emergency.jsx):** The primary step-by-step guidance interface. Renders numbered procedure cards, visual risk indicators, contraindication warnings, and audio playback controls.
- **Emergency Services and Facility Map (Contacts.jsx):** Houses the interactive Leaflet map alongside a localized directory of emergency services, trauma centers, and blood banks.
- **Clinical Protocol and Governance Documentation (About.jsx):** Outlines the project's educational mission, institutional acknowledgments, software versioning, and clinical disclaimers.

9.2 Suggested Component Structure

The user interface is built from modular, highly decoupled components designed for clear visibility and rapid interaction:

- **AiChat.jsx:** A floating conversational component that manages message histories, handles streaming responses from edge functions, and processes speech inputs.
- **ResultCard.jsx:** A card component that displays emergency procedures with visual risk badges, brief summaries, and direct links to step-by-step guides.

- **LanguageToggle.jsx:** A persistent interface control that switches the application between English and Urdu, toggling document directionality (dir="rtl") and updating global language state.
- **Navbar.jsx & Footer.jsx:** Persistent interface frames providing quick navigation, language controls, and direct telephone links to SIEHS 1122 and the Edhi Foundation.

Chapter 10: Backend Design

10.1 Supabase Relational Database Schema and Indexing

The backend data layer uses a normalized relational schema in PostgreSQL 15, optimized for rapid text retrieval and multilingual support.

```
-- Enable trigram extension for fuzzy text matching
CREATE EXTENSION IF NOT EXISTS "pg_trgm";

-- Core emergency conditions table
CREATE TABLE public.conditions (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  slug VARCHAR(64) UNIQUE NOT NULL,
  title_en VARCHAR(128) NOT NULL,
  title_ur VARCHAR(128) NOT NULL,
  risk_level VARCHAR(32) NOT NULL CHECK (risk_level IN ('High', 'Medium to High', 'Low to Medium')),
  summary_en TEXT NOT NULL,
  summary_ur TEXT NOT NULL,
  contraindications_en TEXT[] DEFAULT '{}',
  contraindications_ur TEXT[] DEFAULT '{}',
  created_at TIMESTAMP WITH TIME ZONE DEFAULT timezone('utc'::text, now()) NOT NULL
);

-- Normalized procedural steps table
CREATE TABLE public.condition_steps (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  condition_id UUID NOT NULL REFERENCES public.conditions(id) ON DELETE CASCADE,
  step_number INT NOT NULL,
  instruction_en TEXT NOT NULL,
  instruction_ur TEXT NOT NULL,
  warning_en TEXT,
  warning_ur TEXT,
  CONSTRAINT unique_condition_step UNIQUE (condition_id, step_number)
);

-- Multilingual search synonyms table
CREATE TABLE public.condition_synonyms (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  condition_id UUID NOT NULL REFERENCES public.conditions(id) ON DELETE CASCADE,
  synonym VARCHAR(128) NOT NULL,
  language VARCHAR(8) NOT NULL CHECK (language IN ('en', 'ur', 'roman_ur'))
);

-- Emergency contact directory table
CREATE TABLE public.emergency_contacts (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

```

    service_name VARCHAR(128) NOT NULL,
    phone_number VARCHAR(32) NOT NULL,
    jurisdiction VARCHAR(64) NOT NULL,
    latitude DOUBLE PRECISION,
    longitude DOUBLE PRECISION
);

-- Trigram indexing for high-speed fuzzy search
CREATE INDEX idx_conditions_title_trgm_en ON public.conditions USING gin (title_en
gin_trgm_ops);
CREATE INDEX idx_conditions_title_trgm_ur ON public.conditions USING gin (title_ur
gin_trgm_ops);
CREATE INDEX idx_synonyms_trgm ON public.condition_synonyms USING gin (synonym gin_trgm_ops);

-- Configure Row Level Security (RLS)
ALTER TABLE public.conditions ENABLE ROW LEVEL SECURITY;
ALTER TABLE public.condition_steps ENABLE ROW LEVEL SECURITY;
ALTER TABLE public.condition_synonyms ENABLE ROW LEVEL SECURITY;
ALTER TABLE public.emergency_contacts ENABLE ROW LEVEL SECURITY;

CREATE POLICY "Allow Public Read Access on Conditions" ON public.conditions FOR SELECT USING
(true);
CREATE POLICY "Allow Public Read Access on Steps" ON public.condition_steps FOR SELECT USING
(true);
CREATE POLICY "Allow Public Read Access on Synonyms" ON public.condition_synonyms FOR SELECT
USING (true);
CREATE POLICY "Allow Public Read Access on Contacts" ON public.emergency_contacts FOR SELECT
USING (true);

```

Code 10.1 — Relational schema and RLS policy definitions

10.2 Serverless Edge Functions and AI Orchestration

Dynamic conversational triage is managed by a TypeScript Edge Function (`ai-first-aid/index.ts`) hosted on Supabase's edge network. The function coordinates security, database grounding, and model failover:

Stage	Subsystem / Actor	Operational Mechanics
Request Ingress	Deno Edge Runtime	Validates HTTP POST requests, applies CORS headers, and verifies JSON payload schemas.
Query Expansion	PostgreSQL Client	Matches user input against <code>condition_synonyms</code> to extract matching verified clinical steps.
Context Assembly	Edge Logic Layer	Injects matched clinical procedures and safety guardrails into the LLaMA 3.3 system prompt.
Primary Inference	Groq LPU Client	Executes inference against <code>llama-3.3-70b-versatile</code> with temperature: 0.1 and a 512-token limit.
Automated Failover	Exception Router	Redirects requests to Google Gemini 1.5 Flash if Groq returns HTTP 429 or 5xx errors.
Egress Streaming	Server-Sent Events (SSE)	Streams generated text tokens directly to the

Stage	Subsystem / Actor	Operational Mechanics
		client interface for progressive rendering.

Table 10.1 — Edge Function orchestration pipeline

Chapter 11: Core Business Logic

11.1 AI Medical Prompt Guardrails and Safety Rules

To ensure clinical safety and prevent harmful advice, the conversational assistant runs under strict prompt guardrails:

7. **Mandatory Emergency Service Activation:** The model must open high-risk guidance by instructing the user to contact local emergency services (Rescue 1122 or Edhi 115).
8. **Strict Domain Limitation:** If a user query is not directly related to emergency first aid, physical injury, or acute medical stabilization, the model must refuse to answer and redirect the user.
9. **Prohibition of Invasive Procedures:** The model is strictly forbidden from suggesting prescription medications, invasive surgical techniques, or advanced clinical interventions.
10. **Explicit Contraindication Highlighting:** The model must explicitly warn against harmful layman practices for the identified condition (such as applying ice to burns or offering water to unconscious patients).
11. **Enforced Brevity and Formatting:** Instructions must be structured as numbered, sequential steps beginning with clear action verbs.
12. **Automatic Linguistic Adaptation:** The system must detect whether the user is querying in English or Urdu and generate responses in the matching language.

11.2 Emergency Risk Triage Classification Algorithm

Incoming medical inquiries are categorized into standardized risk tiers to help users quickly assess situation severity:

Risk Tier	Clinical Determination Criteria	Example Conditions	Platform Action
High Risk (Red)	Immediate threat to life; risk of rapid fatal deterioration within minutes.	Cardiac arrest, severe arterial hemorrhage, complete airway obstruction.	Displays prominent red warnings, surfaces immediate 1122 calling buttons, and presents stabilization steps.
Medium to High Risk (Orange)	Severe injury or illness requiring urgent medical evaluation; breathing remains intact.	Second-degree thermal burns, compound fractures, acute heat exhaustion.	Surfaces critical contraindication warnings and displays directions to the nearest emergency

Risk Tier	Clinical Determination Criteria	Example Conditions	Platform Action
			facility.
Low to Medium Risk (Yellow)	Stable injuries without systemic compromise; condition can be managed with standard on-scene first aid.	Minor lacerations, superficial first-degree burns, joint sprains.	Provides standard first-aid steps and guidance on when to seek formal medical evaluation.

Table 11.1 — Risk triage classification tiers

Chapter 12: User Workflow

The platform defines structured, step-by-step procedures for stabilizing patients during common critical emergencies:

Scenario	Entry Trigger	Primary Interventions	Contraindications	Resolution State
Out-of-Hospital Cardiac Arrest	Victim is unresponsive with absent or abnormal (agonal) breathing.	Place on a firm, flat surface. Begin continuous chest compressions at 100–120 BPM at a depth of 5–6 cm. Maintain a 30:2 compression-to-ventilation ratio if trained.	Do NOT place pillows under the head. Do NOT delay compressions to check for pulses repeatedly.	Perfusion maintained until EMS arrives or spontaneous circulation returns.
Thermal Burns and Scalds	Direct exposure to thermal energy, steam, or hot liquids with localized tissue damage.	Cool the burned area immediately under cool, gently running tap water for 10 to 20 minutes. Cover loosely with a clean, sterile dressing.	Do NOT apply ice, butter, cooking oil, toothpaste, or raw flour. Do NOT burst intact blisters.	Thermal transfer arrested; burn surface protected from secondary infection.
Severe Traumatic Hemorrhage	Copious, spurting, or rapidly pooling blood from an open wound.	Apply firm, continuous direct pressure to the wound using a clean cloth or sterile dressing. Elevate the injured site above heart level if no fracture is suspected.	Do NOT remove blood-soaked dressings (layer fresh dressings on top). Do NOT use wire tourniquets.	Hemorrhage managed and stabilized for emergency transport.
Foreign Body Airway Obstruction	Sudden inability to speak, silent coughing, cyanosis, universal choking sign.	Alternate between 5 firm back blows between the shoulder blades and 5 abdominal thrusts (Heimlich maneuver) just above the navel.	Do NOT perform blind finger sweeps in the mouth. Do NOT administer water or liquids to drink.	Airway cleared; victim breathes normally or transitions to CPR if unresponsiveness occurs.

Table 12.1 — Structured emergency protocols

Chapter 13: Public Health Impact, Community Deployment, and Liability Governance

AI LifeSaver addresses longstanding structural challenges in pre-hospital emergency care across developing economies, particularly within high-density urban areas like Karachi. In Pakistan, pre-hospital systems have historically faced significant resource limitations, with public ambulance networks frequently augmented by philanthropic fleets that operate primarily on a transport-first basis.

Community-Level Deployment Settings

- **Academic Campuses and Educational Facilities:** Equips non-medical faculty, administrative staff, and students with guided, interactive emergency tools to improve campus safety.
- **Households and Families:** Provides families with clear, accessible instructions for managing common household emergencies, including childhood choking, scalds, and domestic falls.
- **Community Groups and Disaster Relief:** Serves as a digital support tool for non-governmental organizations conducting first-aid training across underserved communities.

Legal and Operational Boundaries

- **Educational Purpose:** The system functions as an informational reference tool to guide first aid before professional help arrives, explicitly stating it does not provide formal clinical diagnoses.
- **Good Samaritan Protections:** The application's instructions align with basic first-aid principles, encouraging bystanders to act in good faith to stabilize casualties without attempting advanced or invasive medical procedures.
- **Telephony Integration:** By prominently displaying regional emergency numbers (SIEHS 1122, Edhi 115) across all views, the platform reinforces that bystander care is a bridge to, not a replacement for, professional emergency services.

Chapter 14: Non-Functional Requirements

To ensure dependable performance during emergency situations, AI LifeSaver targets specific technical and accessibility benchmarks.

Quality Metric	Performance Target	Architectural Implementation
Inference Latency (TTFT)	Sub-second response on standard 4G connections	Groq LPU hardware running LLaMA 3.3 70B with Server-Sent Events token streaming.
Search Retrieval Latency	Near-instant query round-trip	PostgreSQL Generalized Index Search (gin_trgm_ops) over condition tables and synonyms.
System Availability	High operational uptime	Globally distributed Supabase Edge

Quality Metric	Performance Target	Architectural Implementation
		Functions with automatic failover between Groq and Gemini.
Offline Operability	Full availability of core procedures	Client-side caching using bundled static datasets (firstAid.json) and local PDF rendering via jsPDF.
Data Privacy & Security	Zero Protected Health Information stored	Stateless guest architecture; no personal identity tracking or remote conversation logs.
Accessibility Compliance	WCAG 2.1 Level AA conformance	High-contrast visual themes, readable typography for English and Urdu, and screen-reader accessible controls.

Table 14.1 — Non-functional quality targets

Chapter 15: Expected Outcomes and Empirical Evaluation

Validation tests demonstrate that AI LifeSaver achieves significant performance and reliability improvements over traditional web architectures.

Metric	Traditional Cloud Architecture	AI LifeSaver Configuration	Measured Impact
Time-to-First-Token (TTFT)	Standard Cloud API baseline	Groq LPU Edge Pipeline	Significant reduction in response latency.
Search Retrieval Duration	Unindexed string scan	PostgreSQL GIN trigram index	Substantial improvement in search speed.
Initial Production Bundle	Standard bundle	Vite 7 code-split bundle	Notable reduction in initial payload.
Offline Procedure Availability	Complete interface loss	Available via firstAid.json cache	Complete offline procedural availability.
Medical Hallucination Rate	Unconstrained LLM baseline	Under RAG system guardrails	High clinical output reliability.

Table 15.1 — Benchmark comparison against a traditional cloud baseline

Chapter 16: Technical Horizons and Expansion

- **Computer-Vision Injury Assessment:** Integrating on-device image classification models (via TensorFlow.js) to assist bystanders in categorizing burn depths and wound severity using their smartphone cameras.

- **Direct Emergency Dispatch Integration:** Developing automated Webhook connections with municipal dispatch centers, like SIEHS 1122, to share caller coordinates and preliminary triage data automatically.
- **Peer-to-Peer Mesh Distribution:** Using WebRTC data channels to enable peer-to-peer distribution of emergency guides between mobile devices during municipal network disruptions.

Through its combination of low-latency edge computing, clinical guardrails, and bilingual accessibility, AI LifeSaver provides a practical, scalable framework for reducing pre-hospital mortality in communities where emergency resources are most constrained.

Appendix A: Condition Data Schema and LLM System Prompt Protocol

A.1 Relational Condition and Procedural Seed Specification

The following SQL seed file illustrates the bilingual schema used to store standardized conditions in the database, including procedural steps and contraindication warnings:

```
-- Seed record for acute thermal burns
INSERT INTO public.conditions (
  slug, title_en, title_ur, risk_level,
  summary_en, summary_ur,
  contraindications_en, contraindications_ur
) VALUES (
  'burns',
  'Thermal Burns and Scalds',
  'جھلسنا اور جلنا',
  'Medium to High',
  'Immediate cooling and protection for thermal burns and scald injuries.',
  'حرارت یا گرم مائع سے جلنے کی صورت میں فوری ٹھنڈا کرنے اور محفوظ رکھنے کے اقدامات',
  ARRAY[
    'Do NOT apply ice or freezing water',
    'Do NOT apply butter, toothpaste, cooking oil, or flour',
    'Do NOT burst blisters or tear stuck clothing'
  ],
  ARRAY[
    'برف یا انتہائی ٹھنڈا پانی مت لگائیں',
    'مکھن، ٹوٹو پیسٹ، تیل یا آٹا مت لگائیں',
    'چھالوں کو مت پھوڑیں اور چپکے ہوئے کپڑے مت کھینچیں'
  ]
);

-- Associate procedural steps with the burn record
INSERT INTO public.condition_steps (condition_id, step_number, instruction_en, instruction_ur,
warning_en, warning_ur)
VALUES
(
  (SELECT id FROM public.conditions WHERE slug = 'burns'),
  1,
  'Cool the burn immediately under cool, gently running tap water for 10 to 20 minutes.',
  'پانی بہائیں۔ (نل کا) منٹ کے لیے ٹھنڈا 20 سے 10 فوراً جلنے والی جگہ پر',
  'Use cool running water, not ice or freezing water.',
  'پانی نارمل ٹھنڈا ہونا چاہیے، برف کا پانی برگز استعمال نہ کریں۔'
),
(
  (SELECT id FROM public.conditions WHERE slug = 'burns'),
  2,
  'Carefully remove rings, watches, jewelry, or loose clothing near the burn before swelling begins.',
  'سوچن شروع ہونے سے پہلے جلی ہوئی جگہ کے قریب موجود انگوٹھیاں، گھڑی یا تنگ کپڑے احتیاط سے اتار لیں۔',
  'Do not pull clothing that is stuck directly to the burn.',
  'اگر کپڑا جلد سے چپک گیا ہو تو اسے زبردستی مت کھینچیں۔'
),
(
  (SELECT id FROM public.conditions WHERE slug = 'burns'),
  3,
  'Cover the burn loosely with a clean, dry, sterile dressing or clean plastic wrap.'
```

```
'جلی بوئی جگہ پر صاف، خشک اور جراثیم سے پاک پٹی یا پلاسٹک کی شیٹ ڈھیلی کر کے لپیٹیں۔',
'Do not wrap tightly to avoid constricting tissue as swelling develops.',
'پٹی کو زیادہ سخت مت باندھیں تاکہ سوجن کی صورت میں دباؤ نہ آئے۔'
);
```

Code A.1 — Bilingual seed data for the 'burns' condition record

A.2 Edge Function System Prompt and Guardrail Configuration

The Supabase Edge Function (ai-first-aid/index.ts) enforces the following clinical guardrails, emergency dispatch links, and formatting rules:

```
export const EMERGENCY_AI_SYSTEM_PROMPT = `
You are the AI LifeSaver Emergency Medical Assistant, engineered strictly for bystander first
aid and trauma triage.

OPERATIONAL INVARIANTS:
1. EMERGENCY SERVICES ACTIVATION:
  You must begin every critical response by explicitly instructing the user to call
  regional emergency services: Rescue 1122 (Pakistan SIEHS / Punjab Rescue) or Edhi 115.

2. ABSOLUTE MEDICAL RELEVANCE GUARDRAIL:
  You must evaluate every input. If the query is NOT strictly related to emergency
  first aid, physical injury, trauma, or urgent medical stabilization, you must REFUSE
  to answer. State clearly: "I only provide emergency first-aid assistance. Please
  consult a professional for other inquiries."

3. ZERO GENERATIVE HALLUCINATIONS:
  Never suggest prescription medications, surgical techniques, or invasive medical
  procedures. Limit all instructions to standard Basic Life Support (BLS), wound
  stabilization, and safe patient positioning.

4. CONTRAINDICATION HIGHLIGHTING:
  Always state what NOT to do for the identified emergency condition (e.g., forbidding
  fluids for unconscious individuals, forbidding ice on burns).

5. ACTIONABLE STRUCTURE:
  Structure instructions as numbered, sequential steps. Begin each step with a clear
  action verb. Use bold formatting for critical safety notices.

6. BILINGUAL CAPABILITY:
  Detect the language of the prompt (English or Urdu). Respond in the matching
  language. When responding in Urdu, use natural, accessible vernacular terminology
  suitable for emergency comprehension.
`;
```

Code A.2 — LLM system prompt enforcing clinical guardrails

The technical implementation of AI LifeSaver demonstrates that modern web standards, edge computing, and open-weight language models can work together to address long-standing challenges in

pre-hospital emergency care. By keeping bundle sizes small, providing bilingual support, and enforcing strict clinical guardrails, the platform offers an accessible, reliable tool to help bystanders stabilize casualties during the critical first minutes of medical emergencies.

Disclaimer: This is for informational purposes only. For medical advice or diagnosis, consult a professional.